Sceptre VITA-49 Streaming Format

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The following is the subset of the VITA-49 streaming format used by Sceptre for IQ data streaming. This document describes the format produced using the default settings in Sceptre, specifically with VRL disabled.

General Packet Structure

All packets follow the form:

Field	Start Bit	# Bits	
Packet Type	0	4	Type of packet. 0001 (IF data packet with stream ID) or 0100 (IF context packet)
class_id flag	4	1	Whether or not a class ID is present in the header. Always 0.
trailer_flag	5	1	Whether or not a trailer word is included at the end of the packet. Always 0.
Reserved	6	2	Unused.
TSI	8	2	TimeStamp-Integer format. Always 01 - UTC time
TSF	10	2	TimeStamp-Fractional format Always 10 - Real Time Picoseconds
Packet Counter	12	4	4-bit modulo 16 counter. Counter is independent for IF data packet stream and context packet stream.
Packet Size	16	16	Number of 32-bit words in the packet (including this header).
Stream ID	32	32	Stream Identifier. This allows multiple data streams to be multiplexed on a single packet stream. Stream ID can also be used to associate IF data packets to their corresponding IF context packets.
Integer Timestamp	64	32	Whole part of timestamp (Unix epoch)
Fractional Timestamp	96	64	Fractional part of timestamp (picoseconds)
Payload	160	varaible	Data or context packet payload

*All integers are in big endian byte order.

Note that Sceptre does not include a VRT packet trailer.

IF Context Packet Payload

Sceptre sends context packets periodically to transmit additional metadata needed to interpret the data in the IF data packets. A context packet is sent once per second regardless of whether any of the metadata has changed in order to allow consumers of the stream to connect at any point in time and still be able to initialize the metadata. Sceptre also sends a context packet immediately when one of the parameters in the context packet changes.

Field	Start Bit	# Bits	
Context Indicator Field	0	32	Bit flags that indicate which context fields are present in the packet.
Context Fields	32	variable	Context field values

Sceptre uses the following context fields:

Bit	Field	VITA-49 Section	Included
29	Bandwidth (Usable IF bandwidth)	7.1.5.4	Optional
27	RF Reference Frequency (True RF frequency)	7.1.5.6	Always
23	Gain (Gain reported by radio)	7.1.5.10	Optional
21	Sample Rate	7.1.5.12	Always
15	IF Data Packet Payload Format	7.1.5.18	Always
14	Formatted GPS (Global Positioning System) Geolocation	7.1.5.19	Optional
12	ECEF (Earth-Centered, Earth-Fixed) Ephemeris	7.1.5.21	Optional

Fields marked “Always” will always be included in Sceptre context packets. Fields marked “Optional” will only be included if that information is available. The Context Indicator Field flags will indicate which of the optional fields are present. Context field formatting is described in detail in the specified sections in the VITA-49 standard.

IF Data Packet Payload

IQ data from Sceptre is transmitted in the same sample format of the internal Sceptre stream. If LiveStreaming a RAW stream, the data will be in the same sample format as received from the receiver. Most other streams in Sceptre will be real or complex 32-bit float. The sample format is defined in the context packet "IF Data Packet Payload Format".

All IQ samples are stored in big-endian byte order as prescribed by the VITA-49 standard, so samples may need to be byte swapped if processing the stream on a little-endian processor.

The number of samples in an IF data packet can be inferred from the packet size. Each packet contains 20 bytes of VRT header, so the number of samples is:

```
floor((packet_size * 4 - 20) / bytes_per_sample)
```

The `packet_size` in the VRT header is the number of full 32-bit words. If the sample size is less than 32-bits, the payload is padded out to the next whole 32-bit word.
`bytes_per_sample` is inferred from the "IF Data Packet Payload Format" in the context packet.